

Table of contents

\\Research

<table border="1"><tr><td>Investigators</td></tr></table>	Investigators
Investigators	

<table border="1"><tr><td>Frederick</td></tr></table>	Frederick
Frederick	

<table border="1"><tr><td>GAPS_ONE_4</td></tr></table>	GAPS_ONE_4
GAPS_ONE_4	

<table border="1"><tr><td><table border="1"><tr><td>rslh_ep3d_vaso</td></tr><tr><td>csi_fid</td></tr></table></td></tr></table>	<table border="1"><tr><td>rslh_ep3d_vaso</td></tr><tr><td>csi_fid</td></tr></table>	rslh_ep3d_vaso	csi_fid
<table border="1"><tr><td>rslh_ep3d_vaso</td></tr><tr><td>csi_fid</td></tr></table>	rslh_ep3d_vaso	csi_fid	
rslh_ep3d_vaso			
csi_fid			

\\Research\Investigators\Frederick\GAPS_ONE_4\rslh_ep3d_vaso

TA: 14 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---

Contrast - Common

TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Multi-echo spacing	88.00 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	3130.32 ms
TI 2	9370.32 ms
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Reordering	Linear
------------	--------

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
Base Resolution	84
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.249488 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vso rslh6.0
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1044 Hz/Px
Echo Spacing	1.04 ms
Asymmetric Echo	Off
Turbo Factor	48
Segmentation	1
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1080 us
RF time x BW	18
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
Invert PE	Off
Min. TE w/ PF	On
Expert ramping	Off
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Phase Correction	per Series
EPI ramp factor	1.10
EPI ramp factor 2	1.10
Ramp sampling	1.00 1:max
Slab Scale	0 %
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off
GRAPPA Regularization	5000 /10^6
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Read polarity	Regular RO
VASO Pre-T11 delay	0.00 ms
VASO Pre-T12 delay	0 ms
VASO Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\Research\\Investigators\\Frederick\\GAPS_ONE_4\\csi_fid

TA: 36:24 min Coil Selection: Auto Voxel Size: 50.0×50.0×36.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Vol A >> P	400 mm
Vol R >> L	400 mm
TR	3000.0 ms
TE	2.30 ms
Averages	60

Contrast - Common

TR	3000.0 ms
TE	2.30 ms
Flip Angle	90 deg
Preparation Scans	8
Averages	60
Water Suppression	None

Resolution - Common

FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Scan Res. A >> P	8
Scan Res. R >> L	8
Interpol. Res. A >> P	8
Interpol. Res. R >> L	8
Hamming	Off
Dimension	2D
Phase Encoding	Weighted
Vector Size	512
Normalize	Off

Geometry - Common

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Vol A >> P	400 mm
Vol R >> L	400 mm

Geometry - AutoAlign

Slice Group	1
Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
A >> P	400 mm
R >> L	400 mm
F >> H	36 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 31P	49.891909 MHz
Frequency 1H	123.249488 MHz
? Ref. Amplitude 31P	0.000 V
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	csi_fid
Preparation Scans	8
Delta Frequency	0.00 ppm
Measurements	1
Dimension	2D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	4000 Hz
Acquisition Duration	128 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	31P
TX/RX Delta Frequency	0 Hz
NOE Type	Rectangular
NOE Count	1
NOE Duration	5.00 ms
NOE Flip Angle	90.00 deg
TX Nucleus	1H
TX Delta Frequency	0 Hz
Decoupling Type	Waltz4
Decoupling Duration	1.00 ms
DC Total Duration	50.00 %
DC Pause Fract.	100.00 %
Decoupling Flip Angle	180.00 deg